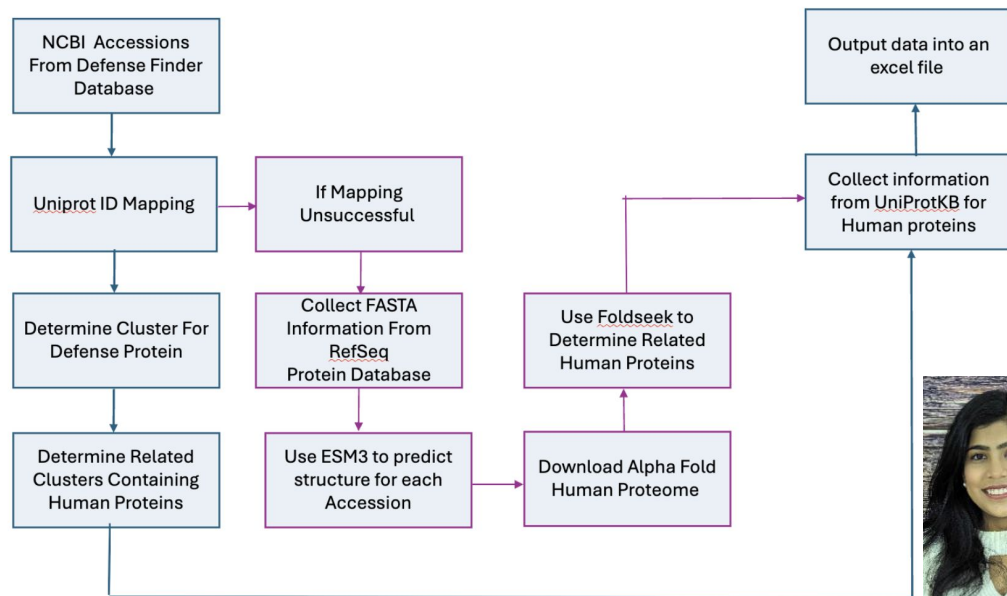


Scaling the Foldseek Homology Pipeline

Background

The workflow maps **RefSeq protein IDs** from bacterial defense systems (e.g., Abi2) to **UniProt accessions**, then leverages the **AlphaFold Database (AFDB) Foldseek clusters** to find structural similarities between these bacterial proteins and human proteins. Finally, it annotates the results with UniProt and domain-level information to identify conserved functional regions across species.



Current Pipeline Bottlenecks & Limitations

- **Compute:** Out-of-Memory (OOM) errors when attempting to load massive datasets.
- **Storage I/O:** High-Performance Computing (HPC) shared file systems suffer from extreme I/O bottlenecks when writing and reading thousands of small flat files (TSV/FASTA) concurrently
- **Orchestration:** Relying on manual shell scripts and rigid, hard-coded Slurm array batches (e.g., 177 batches of 500 accessions) is brittle and difficult to restart upon failure

```
00_get_unique_accessions.ipynb
01_get_uniprot_accessions.py
01_submit_get_uniprot_accessions.sh
02_compare_sequences.py
02_submit_compare_sequences.sh
03_get_uniparc_ids.ipynb
04_process_domain_annotations.ipynb
04_submit_wrangle_swiss_prot_annots.sh
04_wrangle_swiss_prot_annots.py
05_generate_slurm_file.py
06_find_human_foldseek_matches.py
07_run_foldseek-new.py
07_run_foldseek.py
07_submit_run_foldseek.sh
08_add_annotations-optimized-new_metrics.py
08_add_annotations-optimized.py
08_add_annotations.py
08_submit_add_annotations.sh
09_subset_for_griid_genes.ipynb
10_get_fasta_sequences.ipynb
```

Processing at Scale



MinIO Object Storage

MinIO handles massive concurrent read/write requests without locking up the cluster's local disk, making data access fast for distributed Spark workers.



PySpark Processing

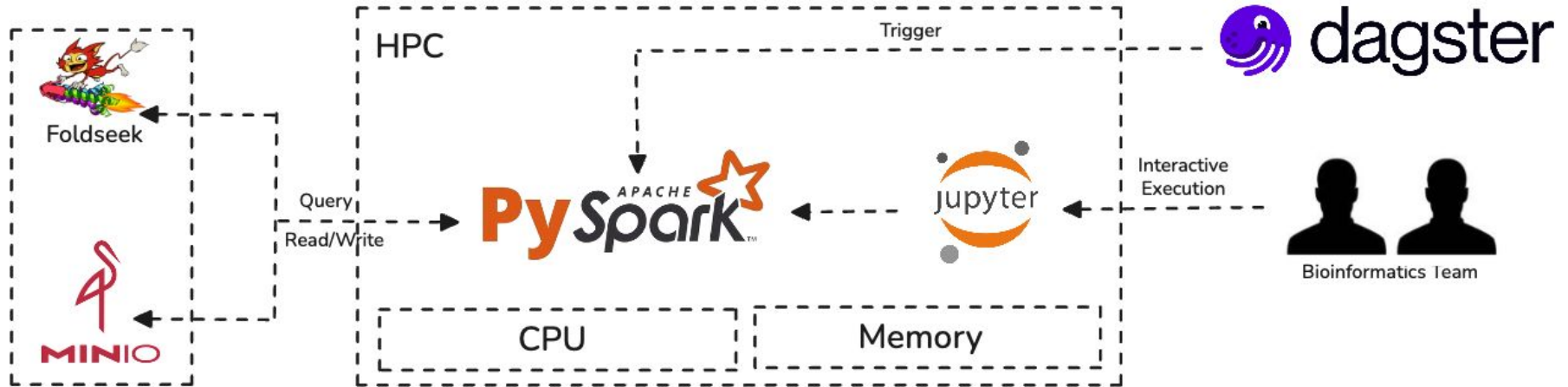
PySpark prevents OOM errors by lazily evaluating execution graphs and distributing the heavy lifting (like joining 88k UniProt accessions against massive AFDB clusters) across available memory pools.



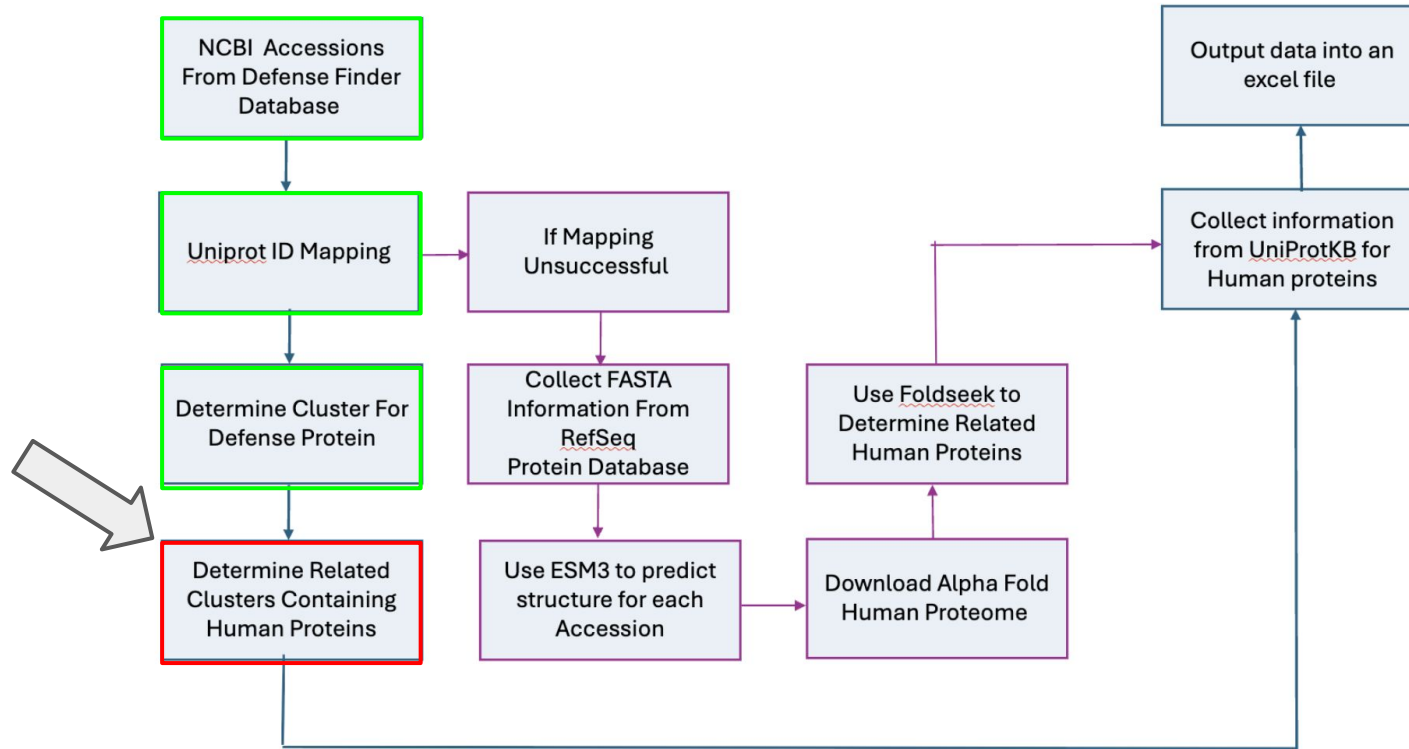
Dagster Orchestration

Replaces manual shell scripts with software-defined assets that automatically track dependencies, lineage, and trigger intelligent recomputation.

Processing at Scale



Current Status



Next Steps

Migrating the Foldseek Clustering Workload

- **The Goal:** Port the AlphaFold Database (AFDB) cluster cross-referencing into PySpark.
- **The Upgrade:** Instead of triggering 177 separate shell batches to find human protein clusters, we will perform a distributed PySpark join. Our clean UniProt accessions will map directly against the AFDB structural cluster data in-memory, bypassing local disk constraints.

Automating Functional & Domain Annotations

- **The Goal:** Integrate UniProtKB and Domain [FT] annotations for both bacterial and human targets.
- **The Upgrade:** Automatically ingest the 8-week UniProt updates. These assets will join the domain metadata onto our structural clusters.